EMMANUEL SAMPSON

Williamsburg, VA • ecsampson03@gmail.com•linkedin.com/in/emmanuel-sampson • github.com/emmanuelcsam

EDUCATION

William & Mary, Williamsburg, VA **Expected May 2026**
Double Major: BS, Physics (Engineering Physics Concentration); BA, Philosophy

TECHNICAL SKILLS

Programming: Python, C++, MATLAB, Java, JavaScript, HTML/CSS
Engineering Software: Autodesk Suite, Fusion360, TinkerCAD, CAD Software, Apex, Makerbot, Jira
Specialized: Computer Vision, AI/ML, Fiber Optics, Signal Processing, RF Engineering, Data Acquisition
Instrumentation: SEM, AFM, Mass Spectrometry, Raman Spectroscopy, Surface Profilers

ENGINEERING EXPERIENCE

Engineer Intern **May 2025 - Present**
WEINERT Fiber Optics Inc, Richmond, VA

- Designed AI-powered computer vision systems in Python/C++ for automated fiber-optic defect detection
- Developed real-time medical temperature monitoring sensors for surgical applications
- Improved transmission efficiency by 30% for medical laser systems

Physics Research Assistant **October 2023 - Present**
Applied Research Center Core Labs, Newport News, VA

- Collaborated with CERN, NIST, and CALTECH on low angle reflectivity experiments
- Assembled TRIFT 4 time-of-flight mass spectrometry instrument
- Developed custom data acquisition systems using Python and LabVIEW
- Operated Hitachi S-4700 FE-SEM, Bruker ICON AFM, and HIROX KH-7700 microscopes

Student Engineer **October 2022 - November 2023**
W&M Makerspace, Williamsburg, VA

- Educated 50+ students on CAD software and fabrication techniques
- Managed electro-optic equipment and 3D printing operations
- Led team projects in mechanical and electrical engineering

Manufacturing Engineer Intern **May 2023 - July 2023**
Build Forward Foundation, Richmond, VA

- Maintained laser cutting and metallurgy equipment
- Developed motorized vehicle prototypes using CAD and rapid prototyping
- Optimized manufacturing processes reducing production time by 20%

PROJECT EXPERIENCE

AI Fiber-Optic Defect Analyzer **May 2025 - Present**

- Built custom microscope with computer vision for automated defect detection
- Achieved 95% accuracy in identifying surface defects using machine learning

Radio-frequency Mechanical Analyzer **January 2024 - March 2024**

- Created RF sensor detecting non-transmitting mechanical items (fans, turbines)
- Implemented signal processing algorithms for frequency analysis

Bird Sensor Technology (Hackathon Winner) **March 2024**

- Developed sensor system detecting bird-window collisions via frequency analysis
- Won William and Mary 2024 Hackathon competing against 30+ teams

Autonomous Medical Drone Routing System **January 2025**

- Developed Python simulation for autonomous medical supply delivery
- Implemented obstacle avoidance algorithms using Blender and optimization techniques

LEADERSHIP & ACHIEVEMENTS

President: Robotics Club (2 yrs), Data Science Society, Tech Executive Association
Technical Leadership: Created and hosted William & Mary's first STEM Symposium
Awards: W&M Hackathon Winner, Catron Scholar, Sharpe Community Scholar